AIA · AGENTIC AI ANALYTICS · METHODOLOGY

# The business-decision study
# A cost model of options

How AIA takes a client's decision from *Zadání* to a report in which every number carries its source and its grade. The method is described generically: it applies to any decision between options whose costs can be modelled.

THE SPINE · EIGHT STEPS

*Zadání* · *Data* · *Rešerše* · *Předpoklady* · *Model* · *Výsledky* · *Sociomapa* · *Osnova a Zpráva*

The client's rule is written at the first step and applied, as written, at the sixth. Every number in the report carries its source and one of seven grades. A method the register does not hold stays provisional, and a recommendation that rests on one is *předběžné*.

**Version 5** · 9 November 2026 · working version, for the product owner and the methodology owner
Replaces version 4 (case-based, 43 pages, 9 November 2026). The method is unchanged; every case reference is removed.
Source: `prototypes/obchodni-model/methodology/` · built by `build.js`
Agrees with the MVP scope: `prototypes/obchodni-model/MVP-SCOPE.md`, version 2

This is design, not product code. Chapter 11 says what exists in AIA today and what does not.

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BEFORE THE CHAPTERS

# What this is

The method AIA follows when a client must choose between options with large sums at stake: a set of sites, suppliers, investments or keeping the current arrangement. It is the measure the MVP is held to.

## What the method produces

A decision between options, judged by a rule the client wrote before any result, on a cost model computed by code, with the uncertainty of every input carried through to the verdict. The report says what the rule chooses, how far that verdict is from changing, what information would change it, and what each option means for finance, operations and the schedule. Two maps show how the options relate across the model's futures and across the client's weightings of the business perspectives; they inform, the rule decides.

Studies without a model (a situation report) and studies built on survey respondents are not this method. The survey lane is frozen (ADR 0046, decision 2).

## How to read it

- Chapter 1 is the spine: the eight steps in one table.
- Chapter 2 is the analytical contract: the defaults every step runs under until the methodologist amends them, and the two release states.
- Chapters 3 to 10 take one step each, in the same six parts: what goes in, what comes out, the checks, what AIA has today, what is new, and what the acceptance test expects.
- Chapter 11 says what is built and what is designed. No code for this study type is on `develop`.
- Chapters 12 to 14 cross the steps: the business perspectives and the two maps; grades, sources and the release gate; governance and acceptance.
- The appendix is a glossary.

Czech words in italics are terms a person sees on AIA's screens or in the report. Words in `code` are files, modules or plan chunks in the repository. "The review" is the product owner's methodology review of 5 November 2026 (ADR 0046 § The methodology review). "The scope" is `MVP-SCOPE.md`; where the two documents disagree, the scope wins. "The owner" is the product owner; "the methodologist" is the methodology owner.

> **Three things this document does not do.** It does not invent a number: the numbers in it are rules, clearly illustrative values, or the toy acceptance fixtures of chapter 14. It does not decide what waits on the owner or the methodologist; chapter 14 lists those items. It does not describe how a screen works: the design boards do that.

CHAPTER 1

# The spine

A study is eight steps. The client's rule is written at the first and applied, as written, at the sixth. Everything between them exists to make that application sound. The navigator shows nine screens: the outline and the report are one step, because the report's structure is built once and printed once.

| Step | Goes in | Comes out | The check |
|---|---|---|---|
| **1** ***Zadání*** | the decision, its options and constraints, who decides and when, the client's files | the rule with its conditions, frozen and dated; interpretations recorded | frozen before any comparative result; every condition has a threshold and a measure |
| **2** ***Data*** | every client file and document | a contract per file; a findings table with a disposition per finding | units, currencies, price years, duplicates, missing values, outliers, quotes that do not cover the same thing |
| **3** ***Rešerše*** | what the files cannot give: public prices, rates, taxes, limits, law | typed public inputs, each with a quote, a source, a date and a state | verified at the source, or marked *prozatímní* |
| **4** ***Předpoklady*** | the contracts, the public inputs, the client's answers | one table of inputs: low, likely, high, distribution, source, grade, guarantor, state | estimated from client data where possible; the scope of every estimate checked; correlations declared with a stability test fixed in advance |
| **5** ***Model*** | the input table, the options, the timeline | the base case and the seeded futures; the same bytes on a rerun | one model for every option; the convergence rule of chapter 2; one change at a time in the extra computations |
| **6** ***Výsledky*** | the futures and the base case | the rule applied as written; what could change the verdict; value of information; finance, operations, schedule | the verdict shown as a boundary with its resolution state; no probability outside the model; information valued for the rule first |
| **7** ***Sociomapa*** | the options' results across the futures, or their weighted scores across scenarios | a map with its mode contract, its relation matrix and what it may claim | links only where the contract allows; a stability test declared in advance; relative apart from absolute |
| **8a** ***Osnova zprávy*** | every claim the report will make | an outline ordered by the argument; each point with its source or method and why it matters | no point without a source; a judgement is marked as one |
| **8b** ***Zpráva*** | the closed outline | the report: summary, chapters, limits, literature, appendices, tables | every number with a grade and a numbered source; a validity review before approval |

## 1.1 The shape of each step chapter

Chapters 3 to 10 use six parts, always in this order, so that two steps can be compared line by line.

| Part | What it answers |
|---|---|
| What goes in | The inputs the step takes, and from which earlier step or from the client. |
| What comes out | The artefact the step produces, as the next step and the report read it. |
| Checks | What must hold before the step is closed. A check is a refusal or a record, never advice. |
| In AIA today | The code on `develop` the step reuses, by module. |

| Part | What it answers |
|---|---|
| New | What has to be built, with the plan chunk that builds it (`plan#CHUNK`; **new** marks a chunk that the scope defines and no plan file holds yet). |
| Acceptance | What the acceptance test (chapter 14) checks at this step. |

## 1.2 Acceptance and waves

Acceptance has two layers. First an independent numerical oracle: toy fixtures with known answers (O-CF for the cash flows, O-INFO for the value of information) and a frozen test configuration, written by an analyst who never writes the model code and signed by the methodologist. Second a frozen test case with planted errors: data faults, costs a brochure hides and a site document reveals, decoys that look like problems and are not. The pipeline must find every planted fact, apply the rule as written, and reproduce its bytes; the case's numbers are not a target (chapter 14).

The build runs in five two-week waves (scope § 7): W1 contracts, oracles, the study shell, the rule freeze and the first data contract; W2 reconciliation, bindings and the cash-flow model; W3 finance, joint uncertainty and the rule with its convergence; W4 information value, the cost partition and both maps; W5 the report, invalidation, acceptance and the independent review. Three build sessions, one analyst session and the methodologist one day a week; 97–139 session-days in all. Week 10 may produce a reviewed *předběžné* report; the calendar does not turn an unresolved comparison into an approved recommendation.

## 1.3 The rules the whole spine keeps

- **The client's rule decides.** It is written before any result and judged as written. Anything else the study computes is evidence for "what could change the conclusion", not for the recommendation.
- **A number is a reference to its origin.** It carries a source, a method, an uncertainty and a validation status. A number the origin does not support is not printed.
- **Unknown is never good.** A missing input keeps a range; a gate that is unknown blocks the recommendation until it is evidenced. A probability is a share of the model's futures, never a claim about the world.
- **Models propose, code computes, a person approves.** A language model drafts text and suggests ranges; every number in the report comes from a program that gives the same bytes on a rerun.
- **A method the register does not hold stays provisional.** Its results print *odhad (metoda čeká na schválení metodikem)*, and a recommendation that rests on it is *předběžné doporučení*: document approval and recommendation eligibility are two states (chapter 2).
- **A comparison is resolved or it is an interval.** The futures double from 20 000 at declared checkpoints until each gate and decisive comparison clears its bound; one that does not prints as unresolved.
- **Changes propagate.** A revised input marks every downstream result and approval stale until someone reviews it. Every extra computation changes one assumption at a time and says so.

CHAPTER 2

# The analytical contract

Seven decisions have to be made before any model code is written, because each of them changes what a number in the report means. This version fixes a default for every one of them. The owner adopted the seven as the baseline on 9 November 2026; the methodologist reviews them on his first day and changes what he disagrees with. Until he does, the defaults bind every step.

## 2.1 The seven defaults

| Decision | Who decides | Default | What breaks if it stays open |
|---|---|---|---|
| **Comparable-cost basis** | methodologist sets the basis; owner approves the business assumption; code enforces it | For every comparable: the original construction estimate, the actual expenditure on the same scope, the exclusions, currency and price year. Overrun = actual like-for-like cost ÷ original estimate − 1. Site-specific items (a rail connection, a bridge, pre-treatment) are added once, never inside the overrun as well. A source that does not say what its percentage measures is clarified before it is read as this ratio (an open owner decision; chapter 6). | The investment, and any gate that reads it, has no defensible basis; a site item can be counted twice. |
| **Complete parameter manifest** | code owns the schema; methodologist approves uncertainty semantics; owner or client approves business assumptions | Every value the model reads is a row: fixed values, ranges, structured tables, taxes and depreciation, tax capacity, timing dependencies, distributions, cross-option dependence, scales and method settings. Each row has unit, period, source locator, the grade attributes, an owner and its dependencies. A bound without a justification blocks every numerical claim that reads it. Counts in the report (inputs, public inputs, judgements) come from the manifest, never from the text. | Hidden defaults escape provenance and invalidation; the text states a count the table does not hold. |
| **Rule freeze and amendments** | owner or client confirms the meaning; code records and enforces | Objective, gates, rates, budget currency and year, the reading of each condition and tie treatment are frozen before any model result. A condition that can be read two ways (an investment ceiling as installed or as total investment, say) has its reading fixed and recorded. Exposure to earlier results is recorded. An amendment is a new revision with reason, approver, a rerun and the old and new outcomes disclosed. | There is no authorised rule to apply; a result can be rationalised by a rule written after it. |
| **Distribution endpoints** | methodologist; owner or client supplies the business judgements | A hard bound defines the support; an elicited quantile is fitted as a quantile, never read as a bound. Reconciled observations keep their joint structure (a comparable's overrun and delay drawn together). Disagreement stays as separate scenarios or an approved mixture. Missing evidence keeps a justified range, never an automatic midpoint or a made-up distribution. | Tail risk acquires a precision nobody stated; truncation hides the cases the gates exist for. |

| Decision | Who decides | Default | What breaks if it stays open |
|---|---|---|---|
| **Numerical convergence** | methodologist approves the error control; code chooses batching and resource limits | Start at 20 000 independent joint futures; double at declared checkpoints, capped at 1,28 million. Each gate gets a repeated-look-valid exact binomial bound; each median comparison an order-statistic bound; each decisive information comparison, which compares expected monetary gains, a repeated-look-valid bound for the *mean* that carries the conditional-estimation error of its answer branches (the estimator, an empirical-Bernstein or t-based confidence sequence for example, is the methodologist's choice). A probability or quantile bound never stands in for a mean bound: two enquiries can agree in how often they pay and differ in sign on average. The total numerical error budget is 5 %, split across comparisons and looks. A bound that overlaps the threshold means *unresolved*, and the report prints the interval. ±0,5 percentage points alone certifies nothing. Medians are compared as medians, not as the median of paired differences. | A seed or a resource cap becomes the decision rule; 0,796 against a threshold of 0,80 on 20 000 draws (exact 95 % interval 0,790 to 0,802) is read as a verdict (chapter 8). |
| **Enquiry updating** | methodologist approves the update; owner or client approves what counts as evidence and what action follows | Every enquiry declares its answer branches, their reliability, cost and duration ranges and the dependencies it shares with other inputs. The joint model is conditioned on the answer, the timetable shifts by the duration, the cost of asking enters, and the rule is reapplied. A letter from an authority does not remove all uncertainty by itself. | The value of information cannot support an action or a deadline; the enquiry table is a wish list. |
| **Release eligibility** | owner sets the policy; methodologist registers methods; code enforces eligibility | Two states, kept apart: **document approval** (this version was reviewed and may leave AIA) and **recommendation eligibility** (the recommendation may be stated as approved). Labelled unregistered estimates are allowed internally and in an independently reviewed *předběžné* client draft. An approved recommendation needs registered methods, the independent review and every decisive comparison resolved (adopted by the owner on 9 November; chapter 13). | "Approved" means two different things to the client and to the methodologist. |

The seven replace an implicit construction-cost uplift, a fixed N of 20 000 as an acceptance criterion and ambiguous release warnings.

## 2.2 The two release states

| State | Granted when | What the report may say |
|---|---|---|
| **Internal draft** | composed by code from a stored run | everything, labelled; never leaves AIA |
| **Approved document, *předběžné*** | the independent review (chapter 14) is recorded with its open items; every number carries its grade; unregistered methods are labelled | a provisional recommendation with its qualifications named; no "doporučujeme" without the qualifier |
| **Approved document, recommendation eligible** | every method the recommendation rests on is in the register; the review's items are closed; every decisive comparison is resolved under the convergence rule | the recommendation as approved investment advice |

Document approval is what `application/report_approval.py` records today, once per stored version. Recommendation eligibility is new: a field of the approval computed by code from the register, the review record and the comparisons' resolution states, never typed.

## 2.3 Who does what in the ten weeks

Three build sessions, one analyst session that never writes model code and freezes the oracle fixtures (chapter 14), and the methodologist one day a week, starting with the fixtures and these seven defaults (the owner, 9 November). The owner and the methodologist decide; a decision that has not arrived blocks the dependent result, not the engineering around it.

CHAPTER 3 · STEP 1

# *Zadání*: the decision and its rule

The study begins by writing down what will be decided, between which options, under which constraints, by whom and when, and by which rule. The rule is frozen before anyone sees a comparison of the options.

## 3.1 What goes in

- The client's brief, as typed or pasted: one field of free text and the files that come with it.
- The decision-maker, the date the decision is taken, and the date the study must be ready.
- The options, each with the client's source for it, and a baseline every option is compared with (often the status quo: keep doing what the client does now).
- The constraints every option must meet: an investment ceiling, a start date, a capacity.
- The client's answers to the study's questions. Where an answer is "we don't know", the input becomes a judgement with a range and says so.

## 3.2 What comes out

- **The decision record:** what is decided, why now, the goal, who decides, by when.
- **The options and the baseline**, named as the client names them, with the source of each.
- **The rule**, frozen and dated, in the client's words: the measure it reads, the choice it makes, and each condition with its threshold. A typical rule: choose the option with the highest median saving against the baseline, measured as the present value of the cost of supply over the horizon at a stated rate, if the probability that it beats the baseline, the probability of a loss above a stated sum and the probability of starting by a stated date each meet their thresholds, and its investment stays under the ceiling; else recommend nothing and name the assumption most worth checking.
- **Interpretation records.** Where a condition does not say how it is measured, the study writes its reading down, the alternative reading, what each would change, and who confirms it. A ceiling with no probability attached, for example, can be judged on the base case or "in every future"; the record says which, prices the other, and names the person who confirms.
- **The views the client asked for beside the rule:** after-tax values, the verdict at other rates. They are shown; the rule does not read them.
- **The analyses the report will carry** and the perspectives that apply (chapter 12).

## 3.3 Checks

- The rule exists before the first comparative result, and the freeze is dated and attributed. Reading the client's files at intake is feasibility work and is recorded as such.
- Every condition has a measure the model can compute and a threshold. A condition without one gets an interpretation record, never a silent default.
- An amendment of the rule after the freeze is approved by the decision-maker, dated, and printed in the report.
- Every option is described comparably: the same scope, the same horizon, the same discount rate.
- The brief is Class B material by rule (`domain/material_rules.py`); a language model may read it. A client's dataset is Class A and is not sent to a model.

## 3.4 In AIA today

- The study's working copy with revision-checked saves and the operation log (`application/working_copy.py`, `domain/edit_ops.py`, `infrastructure/study_workspace_repository.py`), and immutable Design Revisions (`domain/design.py`, `infrastructure/study_design_repository.py`). The frozen rule is a revision of this kind.
- The Brief stage of the research workbench (`apps/web/src/components/rehome/research/`, `BriefStep`) and its materials (`domain/brief_materials.py`), reused as the screen the brief is typed into.
- The Lead, which triages a brief and proposes edits a person accepts (`application/research_lead.py`, ADR 0026, ADR 0041).
- The approval ledger with its closed set of subjects (`domain/approval_ledger.py`): the rule's freeze and every amendment become ledger rows.
- Material classes by rule (`domain/material_rules.py`, ADR 0008).

## 3.5 New

- The business *Zadání* inside the one merged study, with no new study kind (`cost-model-study#CP3`): the eight parts (`decision-brief#DB0`), triage as a Lead turn (`#DB2`), the decision core and the rule (`#DB3`), readiness (`#DB4`), the hand-off naming the files each step takes (`#DB5`).
- The rule as a typed object of n conditions, each with its own threshold and, where stated, a probability, judged per option at the stated rate and two alternates, frozen as a Design Revision (`decision-brief#DB3b`, **new**, extending `cost-model-study#CR7`, which judges one quantity against one threshold). The screen shows each condition with its threshold and what it reads, the gates that no saving outweighs, and the lock line: frozen on, amendments, interpretations, awaiting confirmation.
- The interpretation record a named person confirms: what the rule does not say, the chosen reading, the other reading and its effect, who wrote it and who confirms, where the report prints it (`decision-brief#DB3b`).
- The model's parts as data: the target quantity, the baseline, the futures, the shared timeline, what is in and outside the model, the investment ceiling as a gate, the result views the model will feed, and the perspectives that have components in the study.
- The perspectives chosen for the study (chapter 12), each with its leaves, in *Zadání* rather than in the model.

## 3.6 Acceptance

- The test case's rule is captured with every condition and its "recommend nothing" clause, dated before any result.
- Every condition that can be read two ways has an interpretation record naming who confirms it.
- The baseline is the status quo at its stated terms, not one of the options.
- The study asks the client only what the test case answers; anything else is a judgement with a range, labelled *úsudek*.

CHAPTER 4 · STEP 2

# *Data*: the client's files under contract

Every client file is read, checked and bound by a contract before a number from it reaches the model. The checks produce findings, and every finding gets a disposition the report can print.

## 4.1 What goes in

- The client's data files: purchases, prices, volumes, quotes, wages, measurements. A file that holds respondents' answers belongs to a survey study and is not read here.
- The client's documents: briefs, offers, memos, letters. Each number taken from one is cited with its page.
- The decision record from step 1, which says what each file is for.

## 4.2 What comes out

- **A data contract per file:** which columns are used and how; units, currency and price year per column; which columns are personal and never leave the analysis copy; the period covered (`client-data-in-analysis` chunk 2).
- **A quality report per file**, accepted by a person before use: missing, duplicate and out-of-range counts per used column over every row (`client-data-in-analysis` 2a).
- **The findings table with dispositions.** Every finding of the checks, with one of four dispositions: used; corrected, with the reason; left out; taken over as an assumption with a range. The findings that change the result or its reading go into the report; the rest stay in a separate data-check document the report names.
- **A document register:** each document by the name the report cites it under and its file name.

## 4.3 Checks

| Check | What it looks for | Typical instance |
|---|---|---|
| Units | a row whose unit differs from its column's | one month's volume in kilograms in a column of tonnes |
| Currencies | mixed currencies in one column; a currency column that contradicts the header | one purchase row priced in another currency |
| Price years | amounts in different years' prices without a year | a comparable project with no price year |
| Duplicates | the same entity twice | one comparable entered twice, once with a rounded figure |
| Missing values | cells the contract expects and does not find | a price series that stops before the horizon |
| Outliers | a value outside the range the file itself declares | a capacity recorded in another unit, far outside the file's range |
| Scope of a quote | quotes that do not cover the same thing | a cheaper quote that covers only part of the service; a rate that assumes infrastructure the site does not have |
| File against brief | a number the brief states and the file contradicts | a price the brief rounds that the purchase file shows rising |

| Check | What it looks for | Typical instance |
|---|---|---|
| Personal data | a column holding a person's identifier | a named contact or an employee number |

A contract that names a column the file does not hold, or a quality report the person has not accepted, blocks the file from the model.

## 4.4 In AIA today

- A client's sources kept once per client and SHA-256, read by a format reader into segments with locators (`domain/client_sources.py`, `infrastructure/source_reading.py`, `infrastructure/client_source_repository.py`): PDF by page and table row, XLSX and CSV by sheet and row, DOCX by paragraph.
- The dataset preview: per column the header, kind, fill, distinct values and samples, with personal columns detected and masked (`domain/client_datasets.py`; patterns pinned in `fixtures/client_datasets/personal.json`).
- The intake sheet, *Přidat podklady*: each file kept and read at once, one knowledge item per file (`apps/web/src/components/aia/clients/IntakeSheet`).
- Document text with ZIP bounds and a bounded table reader (`infrastructure/document_text.py`).
- The evidence policy for client data: client-reported evidence under an accepted contract (ADR 0027, ADR 0037; `client-data-in-analysis` 8a built).

## 4.5 New

- The data contract as a type and the route a person accepts it through (`client-data-in-analysis#2`), the quality report (#2a), the analysis copy of named columns and its accessor (#3, #3a, #3b), admission as *z dat klienta* (#5), the contract accepted on the Data step (#10). The screen shows the contract per column with its check note, the quality report with a disposition per row, what the model gets from the file (summaries, never rows), and the findings with those that change the result marked.
- A number from a document proposed with its page and checked literally on that page (`client-data-in-analysis#11`); an estimate from client rows as a Metodika method (#12).
- Declared conversions (unit, currency at a dated rate, price year), each a finding with its disposition, and the findings table with the four dispositions *použito / opraveno / vynecháno / převzato jako předpoklad* (`client-data-in-analysis#13`, **new**). `cost-model-study#CR1` covers reconciliation, period coverage, currency and robust outliers, accepted on record.
- The document register with cited names (part of `decision-brief#DB5`'s hand-off).

## 4.6 Acceptance

The frozen test case plants faults of every kind in the table above. A study that misses one has a gap in its checks, whatever its verdict.

- Each planted fault is found and has a disposition; a corrected one propagates to every result that read it.
- Where one row can be read two ways (a price alone or a price plus freight, say), the study says which reading it prints and why.
- A quote that covers less than its competitors is compared on the same scope, or not at all.
- Where a table and the text disagree about a reading, the disagreement is found and resolved before the report prints either.

CHAPTER 5 · STEP 3

# *Rešerše*: public inputs with a state

What the client's files cannot give, public sources give: prices, rates, taxes, limits and law. Each is a typed input with a quote, a source and a date, and it stays provisional until someone has verified it at the source.

## 5.1 What goes in

- The list of public inputs the model needs, from the decision record and the contracts: market prices and their outlooks, regulatory thresholds and limit values, fees, wages and employer contributions, tax rates, depreciation and the nature of each incentive, energy price outlooks, exchange rates.
- The client's data posture (*Pravidla*, `domain/client_rules.py`) and the study's budget, which bound what a priced route may spend.
- Findings that frame the problem: the literature the report cites, found and quoted the same way.

## 5.2 What comes out

- **Typed public inputs:** a value with its unit, the quote it was read from, the source and its date, the kind of source (*Úřední / Oborový / Ostatní*), the grade (*veřejný zdroj* or *právní předpis*) and the state: *ověřeno* once checked at the source, else *prozatímní*.
- **Findings by analysis:** PESTEL, Porter or benchmark findings as proposals a person accepts, each a quote in a captured source.
- **The literature review** as the report prints it: for each work, what it says and what follows for this report.
- **The state line for the report's first page:** how many inputs are public, how many unverified, and whether the unverified ones could change the verdict.

## 5.3 Checks

- A finding is a quote in a captured source, or it is nothing (ADR 0017). A value without a quote and a date is not an input.
- A query's data class is inherited from what it was written from and never lowered by its words (ADR 0017, `domain/deep_research/classification.py`). A client's dataset never leaves as a query.
- A provision is cited; no legal advice is given.
- An unverified input is marked *prozatímní* in the input table, in the chapter that uses it and on page 1. Verification at the source is a dated act by a named person and changes the state, nothing else.
- Every unverified input that enters the model says whether its range could change the verdict (chapter 8).
- An incentive is read by its nature: a tax relief does not change a pre-tax figure; only a cash grant reduces the investment.

## 5.4 In AIA today

- Deep Research, frozen since #166: plan, investigate, merge, verify, synthesize, publish; grounding of every quote in a content-addressed snapshot of the same track (`domain/deep_research/`, `grounding.py`, `web.py`, `sources.py`); the sealed bundle.

- The one way a query or URL leaves: `application/web_retrieval.py` `RetrievalGate` (classify, egress, metering, reserve, journal, call, journal). Live routes: Czech and English Wikipedia, priced Brave, Common Crawl (`aia_executors.deep_research_runtime`).
- Design Research and Interpretation Research as the two purposes; a finding never writes a design or a deterministic artefact (ADR 0021).
- A captured source a person checked becomes a SOURCE knowledge item, *Ověřeno, přidat zdroj* (`application/verified_sources.py`).
- The Lead's web lookup, accepted and not built (ADR 0040).

## 5.5 New

- Typed inputs (value, unit, period, range, source, as-of date) extracted by code from accepted quotes and from structured routes, with the state *prozatímní / ověřeno u pramene* (`deep-research-v2#DRV1` the typed input, `#DRV2` the extractor, `#DRV3` the extraction step); credibility tiers *Úřední / Oborový / Ostatní* (`#DRV5`); as a provenance of a model input, `cost-model-study#CP1`'s `public_source`. Verification is a step with three outcomes: Deep Research finds and cites (*prozatímní*); the researcher opens the source and records the date and their name (*ověřeno u pramene*); what no source holds stays a judgement with a guarantor.
- Structured public routes: SDMX for Eurostat and the ECB (`public-data-routes#PDR1`), ČNB rates (`#PDR2`).
- A legal point as a verbatim provision that becomes a parameter (tax rate, depreciation, contributions): e-Sbírka and EUR-Lex routes, the legal point, legal parameters (`legal-context#LGL1`, `#LGL2`, `#LGL3`).
- The first-page state line composed from the input table's states (part of the report recipe, `research-findings-stage#FND13`, chapter 10).
- Findings by analysis (PESTEL, Porter, benchmark) as proposals, each a quote.

## 5.6 Acceptance

- Every public value in the model carries a quote, a source, a date and a state; none is invented.
- A regulatory threshold the test case puts an option above or below is read from its provision, and the cost it brings is in the model for every option it applies to.
- Incentives are read by their nature; subtracting a tax relief from the investment is a misreading and fails.
- Where a public range is a decisive input, the result says so; where the plan of action depends on verifying it, the verification has a date.

CHAPTER 6 · STEP 4

# *Předpoklady*: one table of inputs

Every input of the model is one row of one table: its low, likely and high value, its distribution, its source, its grade, the person who stands behind it and its state. The table is the contract between the data and the model, and the report prints it whole.

## 6.1 What goes in

- The data contracts and the findings table from step 2; the typed public inputs from step 3; the client's answers from step 1.
- The comparable cases the client holds, from which ranges can be estimated rather than judged: comparable projects with their cost overrun, their delay and what each built beyond its core scope.
- The scope statement of every estimate the client supplies: what a contractor's figure includes and excludes, with the page that says so.

## 6.2 What comes out

**The input table.** Its columns are the method:

| Column | Rule |
|---|---|
| *Vstup, jednotka* | one input, one unit; a fixed parameter is a row like any other |
| *Dolní · nejspíš · horní* | three values; equal for a fixed value. What the ends mean is written down for every decisive input, and the meaning decides how they become a distribution (chapter 2): a hard bound (contractual, physical) defines the support; an elicited quantile (a 90 % interval, a credible worst case) is fitted as a quantile and never read as a bound; a missing fact keeps a justified range, never an automatic midpoint |
| *Rozdělení* | PERT by default for a range a person stands behind, its endpoints read as the row says; the estimate's own distribution where it comes from data; an empirical draw from comparable cases that keeps their joint structure; yes/no with a probability for an event; disagreement kept as separate scenarios or an approved mixture; a fixed value (*pevná hodnota*) |
| *Zdroj* | the file and row, the document and page, the public source and date, or "not in the files" |
| *Stupeň* | one of the grades of chapter 13: *z dat klienta*, *odhad*, *úsudek*, *veřejný zdroj*, *právní předpis* |
| *Stojí za ním* | the guarantor: a named role in the client (finance, purchasing, environment and safety) or the author. Every assumption has an owner |
| *Stav* | *ověřeno* (from client data), *prozatímní* (public, unverified) or *úsudek* (the client has no answer) |

Beside the table: **the correlation record** (which inputs move together, with what coefficient, why, and the stability test their judgement will be put to), **the scope check** of every supplied estimate against each option's documents with the exclusions priced, and **the stress cases** outside the ranges, kept apart from the futures.

## 6.3 Checks

- Where the client's data allows an estimate, the range comes from the data, not from a judgement; the row says which. Overrun and delay come from the comparables, not from an estimator's ± band.
- Two things that happened together in the data are drawn together in the model. Where overrun and delay move together across the comparables, each future draws one real comparable and takes both.

- Every supplied estimate is checked for scope against each option's documents, and each exclusion is priced or stated as missing. A study that adds nothing to a contractor's figure has not done the check.
- **The comparables' cost basis is reconciled before their overrun is used** (chapter 2). A file can report totals that include off-site infrastructure and overruns against the estimate at decision; applying that overrun to an option's construction estimate and then adding its site items can count them twice. The overrun is defined as actual like-for-like cost over the original estimate, the exclusions are added once, and an unclear source is clarified first (an open owner decision). The cohort is checked for comparability: a declared size range, duplicates removed, outliers treated by a declared rule.
- **Every value the model reads is a row of the manifest** (chapter 2): taxes and depreciation, the form of each incentive, regulatory limits and the method settings included. The counts the report prints come from it.
- A decisive input has an uncertainty type: natural variation, parameter, expert disagreement, model structure, or a severe event outside the range; the last goes to a stress test, not into the distribution.
- Experts who disagree from the same documents are a mixture that does not narrow the range; the report says they differ in reading, not in facts.
- A correlation that is a judgement is declared as one, with the alternative settings it will be tested under, before anything is computed.
- No input the files do not hold is given a point value. It gets a range and the grade *úsudek*.

## 6.4 In AIA today

- Nothing holds an input table yet. The nearest ancestors are the simulation core's frozen WorldModel with reject-not-clip validation (`domain/simulation/`) and the research template's readiness rules (`domain/research_design.py`), which show the pattern: a typed object, validated, pinned to a run.
- Guarantors and audit: the approval ledger (`domain/approval_ledger.py`) and the audit trail every admin route writes.
- Methodology as data the methodologist owns: versions, activation and bounds per editable field (`domain/methodology/`, ADR 0023). The distribution defaults and the stability thresholds belong there.

## 6.5 New

- The input as a platform type with provenance (`cost-model-study#CP1`: `client_dataset`, `stated`, `public_source`) and low / base / high with a guarantor on every input (`#CR2`); the input table as the stage *Předpoklady*. The screen filters by state and marks the decisive rows; shows for each decisive input its uncertainty type, the meaning of its ends and the reason; the estimate from comparables; the scope check as a table of excluded items (where, by what document, price, months, grade); and the correlations as judgement with the stability test and its settings, dated before the computation.
- Distributions and the correlation record: PERT, declared rank correlations through a Gaussian copula with a validated rank-to-latent conversion and a valid latent matrix (the methodologist's item; the contract recommends it over sample-wide Iman–Conover reordering so the numerical error contract matches the sampler), a non-PSD matrix refused (`#CR3`); empirical joint draws, one row of a client table per future carrying several inputs together, the table pinned by SHA-256 (`#CR3b`, **new**); yes/no events with a probability, fixed and uniform values; people who disagree a mixture (`#CR3`).
- The estimate from comparables as a Metodika method (`client-data-in-analysis#12`).
- The scope check: what an estimate's document excludes against what each option's documents require, each exclusion an input with its range (`cost-model-study#CR1b`, **new**).

- The stability test declared before computing, as the futures map's chunk defines it (`sociomapa-systemu#SYS9b`, chapter 9).
- The state column and its three values, read from steps 2 and 3 (`deep-research-v2#DRV1`'s state for public inputs).

### 6.6 Acceptance

The frozen test case hides costs that a sales document leaves out and a site document reveals. The study must find and price each one.

- Every planted hidden cost (a treatment step a local limit requires, compensation a permitting authority may impose, infrastructure the investor must build) is found from the document that reveals it and priced with a range and a delay.
- Overrun comes from the comparables, with duplicates removed first, not from the estimator's band.
- The decoys are left alone: a past exceedance long since resolved, a limit the option stays under, a minor item well inside the ranges.
- A recurring cost named only in a side document is in the model.
- Decisive distributions, their dependence and the stress cases are each justified in the table.

CHAPTER 7 · STEP 5

# *Model*: one computation for every option

The model computes the measure the rule reads for every option in the same futures. It is code, pinned by version and seed, and a rerun gives the same bytes. Nothing in it is a forecast or a legal opinion.

## 7.1 What goes in

- The input table of step 4, frozen as a revision; the options and the baseline of step 1; the rule's measure.
- The timeline every option shares: the decision date, permitting with each option's own extensions, construction with the drawn comparable's delay, ramp-up below full capacity for its first months.
- The seed and the number of futures, or a stopping rule that sets the number.

## 7.2 What comes out

- **The base case:** one computation with every input at its likely value. It compares the options and feeds the sensitivity analyses; it is not an estimate of what the client will save.
- **The futures:** N seeded draws of all inputs at once, with their correlations, each future computing the baseline and every option. From them: the median, the 90 % interval, and every probability as a share of futures.
- **The extra computations** on the same model and the same futures, each changing one assumption: the decision boundary over an assumption's range, delay of the decision, hiring delay, alternative terms, correlation settings.
- **The run record:** model type and version, input revision, seed, N, the methods pinned, the artefacts with their SHA-256.

## 7.3 Checks

- Every option is computed in the same futures by the same model; the baseline is one of them.
- The same inputs and seed give the same bytes. This verifies the implementation, not the model's fitness for the decision (chapter 14).
- The number of futures follows the convergence rule of chapter 2: 20 000 independent joint futures, doubled at declared checkpoints up to 1,28 million, until every gate and decisive comparison clears its repeated-look-valid bound; a comparison that has not cleared prints as unresolved. A fixed N and a ±0,5 pp margin alone certify nothing; 20 000 is the starting point, not the criterion.
- Cash flows are discounted at the rule's rate, nominal or real as declared, with the convention named (mid-year or end-year; the inflation assumption and its state).
- Tax is inside the model as a fixed parameter per jurisdiction, so the after-tax view agrees with every other view. The rule reads the pre-tax figure unless it says otherwise.
- What is outside the model is listed where the model is described, not discovered in the limits: transfer prices, hiring time in the schedule, freight the delivered price already holds, the model's time step.
- An extra computation changes one assumption; combinations that are not computed are said not to be.

## 7.4 In AIA today

- Runs as workflows on the worker, with lease-fenced steps, artefacts, provenance and dependency edges, each artefact pinned by SHA-256 (`apps/worker`, `infrastructure/artifact_repository.py`, `domain/workflow.py`, `domain/pipeline.py` fingerprints).
- A run pins the organization's methodology in force and records it in its identity (`domain/methodology/pins.py`, `application/research.py`).
- The deterministic simulation core: a frozen WorldModel, reject-not-clip validation, seeded scenarios and frozen results (`domain/simulation/`). Its discipline is the model's; its content is not.
- Seeded synthetic data from `random.Random(seed)` and the Sociomap golden fixtures pinned by SHA-256: the rerun-equals-bytes practice (`packages/aia_core/tests/fixtures/sociomap/`).
- Budgets and cost ceilings per run, which a model computed by code does not need (a CALCULATED run has a known 0,00 USD ceiling, `cost-model-study` CP5).

## 7.5 New

- The platform: the model-type contract and the determinism harness (`cost-model-study#CP2`), the kind and the working copy (`#CP3`), the run with inputs frozen by artefact id (`#CP5`), the API (`#CP6`), the ESTIMATED grade with seed and method pinned and the Metodika module of defaults (`#CP10`). The screen shows the shared timeline as one drawing and the items of the cost of supply with their kind and base formula.
- **A model type of its own, `site_cost_of_supply`** (`cost-model-study#CM8`, **new**): a timeline per option whose dates are inputs, purchases, operations, residual value, tax as a fixed parameter, yearly cash flows, NPV before and after tax at a stated rate, the decision, finance and schedule tables as its results, and the convergence rule of chapter 2. The planned type *scénáře nákladů* (`#CM1`–`#CM7`) has no timeline and no bought baseline, so this is a second type rather than an extension.
- Monte Carlo with the input table's distributions and correlations (`#CR3`), the joint draw from comparables (`#CR3b`), PCG64 under the run's seed.
- The cash-flow schedule, NPV and discounted payback, and IRR by a stated root rule (`#CM7`; `quantitative-families#FIN1`, `#FIN2`).
- The extra computations as declared operations on a stored run, each one change at a time and said so: the boundary, delay and hiring tables, alternative terms and correlation recomputations (`cost-model-study#CR7b`, **new**, chapter 8).

## 7.6 Acceptance

- First the oracle fixture O-CF (chapter 14): two years, 10 % discount, present costs 173,554 / 154,132 / 158,843 and a 12,1 receipt in year 2 worth exactly 10. Then the test case's seed and 20 000 futures as the first checkpoint; the same seed gives the same result.
- A late start is bought at the spot price for the months between the end of the current terms and the plant's start.
- Tax reliefs stay out of the pre-tax figure the rule reads; after-tax values use each jurisdiction's rate.
- Every option shares the timeline from the decision date; an option's extra months (compensation, infrastructure, a permit) enter its own schedule.
- The options are comparable, the material costs and dependencies are in the model, and a revised input invalidates the run's results and approvals.

CHAPTER 8 · STEP 6

# *Výsledky*: the rule applied, and what could change it

The results answer four questions in order: what the rule chooses, how far the verdict is from flipping, what information would change it, and what the finance, operations and schedule of each option look like. The first question is the recommendation; the rest is evidence about it.

## 8.1 What goes in

The futures and the base case of step 5; the rule of step 1; the extra computations declared in step 5.

## 8.2 What comes out

Five views, each a set of tables the report reuses:

| View | Content |
|---|---|
| *Rozhodnutí* | the rule table per option (the measure's median and 90 % interval, each condition's value against its threshold, pass or fail), at the rule's rate and the client's other rates; the **decision boundary**: the rule's choice as one decisive assumption moves over its range; delay tables; stress tests; the tornado; break-even values; alternative terms; the correlation check |
| *Hodnota informace* | two measures, in this order: **the value for the rule** (what knowing an answer in advance and applying the rule again would bring, averaged over the possible answers), then **EVPPI** (the gain in mean saving from knowing one input); then **the enquiry table**: what would be learned, the output, the duration, the cost, and the consequence of waiting |
| *Finance* | pre- and post-tax NPV at the rates, IRR, payback, investment by item with the probability of staying under the ceiling; incentives by their nature |
| *Provoz* | cost per unit by category in a full year of operation, base case, each column named as its perspective leaf (chapter 12) |
| *Harmonogram* | the start-date distribution per option against the deadline (median, 10th–90th percentile), P(on time), the effect of each month of delay, what lengthens each tail, and decision delay × hiring delay as one grid whose uncomputed cells say so |

## 8.3 Checks

- The rule is applied exactly as written, condition by condition; the pass column is computed, never typed. The report prints the value of every condition against its threshold, pass and fail alike.
- The verdict is given as a boundary, not a point: for the one or two assumptions the verdict rests on, the table of what the rule chooses at each value, each cell carrying its resolution state under the convergence rule. A probability of 0,796 against a threshold of 0,80 on 20 000 independent draws has an exact 95 % binomial interval of 0,790 to 0,802, which covers 0,80 before any allowance for repeated looks or several comparisons widens it further; that cell is unresolved until the draws double enough to clear it, and the report says so instead of naming a winner. A verdict with no boundary table has not been tested.
- A probability is a share of the model's futures under the inputs that produced it; every probability says which assumption it stands on. The report claims no probability for anything the model does not compute.

- The value of information is computed for the rule first and for the mean second, through declared answer branches: each enquiry states its possible answers, their reliability, cost, duration and the inputs it shares with others; the joint model is conditioned on the answer, the timetable shifts, the cost of asking enters, and the rule is reapplied (chapter 2). Neither measure is a budget for the enquiry. The rule's value and the expected-benefit EVPPI are different numbers and the report names which it prints. Both are differences of expected gains, so a comparison between enquiries, or against "nic", is resolved by the mean bound of chapter 2's convergence row (a repeated-look-valid confidence sequence for the mean, carrying the branches' conditional-estimation error), never by a probability or quantile bound; an enquiry whose mean bound covers zero stays unresolved and is not recommended.
- A median is not a mean: the rule's choice of statistic is kept, and the report says what the right tail does to the other.
- The base case and the futures may disagree (break-even in the base case, a loss in the median); the report explains why when they do.
- Every extra computation names the one assumption it changes.

## 8.4 In AIA today

- A number reaches a result only as an admitted claim (`domain/evidence/admission.py`, `application/admitted_results.py`); suppression is decided on the server and a suppressed row prints its reasons, never a number. The pattern holds for model results: a cell the method does not support is not printed.
- The results screens of a research run and their reading in words (`ExecutionSteps.tsx`, `lib/result-reading.ts`): unknown is never ordered, an interval missing means no order.
- The report's print grades and the evidence ledger (`domain/report/evidence.py`), the per-question tables as exports (`domain/report/table_export.py`, RD12).
- The budget line and spend reading as the pattern for "unknown never zero" (`lib/budget-line.ts`, `lib/run-spend.ts`).

## 8.5 New

- The rule judged as written (`cost-model-study#CR7`) and, new, the rule at three rates, the decision boundary over one input's range, delay and hiring tables, alternative-terms and correlation recomputations, each one change at a time and said so (`#CR7b`, **new**); the method on screen (`#CR8`); grades ESTIMATED for everything from the futures and CALCULATED for the base case (`#CP8`, `#CP10`). Each result section names the perspective leaf it feeds (chapter 12).
- Value of information for the rule: the rule re-applied per declared answer branch, the average of what it then picks against today's "nic", the enquiry's cost and duration inside the comparison, and the enquiry table with its effect on the deadline (`#CR6b`, **new**); EVPPI beside it (`#CR6`), its estimator's bias and precision checked on the oracle fixture O-INFO (chapter 14).
- Break-even and one-way sensitivity (`#CM2`), global sensitivity beside the tornado (`#CR4`), three chart kinds the results need and the model lacks: the absolute stacked bar, the tornado, the line with marked points (`#CP8`).
- Finance: scenarios on NPV and IRR (`quantitative-families#FIN3`), tax relief against grant as legal parameters (`legal-context#LGL3`).
- Operations: cost per unit by category in a stated year; *Harmonogram*, the start-date distribution per option against the deadline, P(*včas*) and the months-of-delay table (`cost-model-study#CM9`, **new**).
- Accounting and market views are parked: they read no leaf of the cost partition.

## 8.6 Acceptance

The test is not a target number. A report is judged by whether it found the planted facts and applied the rule honestly. Where the test case can be read two defensible ways, each reading is frozen as its own configuration of the test (chapter 14), and each configuration's numbers are a target for itself, never for the other.

- The rule is applied as written, with every condition's value printed.
- The verdict is given with its boundary on the assumption it rests on, and each boundary cell with its resolution state.
- The assumption most worth checking is named with its value for the rule.
- The hidden costs and the overrun from the comparables are visible in the investment and the schedule.
- An option that a sales document favours and the files rule out is not recommended.
- The rule's choice is kept apart from supplementary scoring; every enquiry states its effect on the deadline; the rule table, the summary and the boundary table print the same numbers.

CHAPTER 9 · STEP 7

# *Sociomapa*: the futures map

A Sociomap places the options as points in a landscape: two options lie close when the same observers rate them alike, and the height shows the mean rating. In a business-decision study the observers are the model's futures. The map is evidence about the structure of the options' risks, never about the recommendation.

## 9.1 What goes in

- The futures of step 5 as a table: one row per future, one column per option, the cell the measure the rule reads (the saving against the baseline in that future).
- The mode contract for futures (mode *Scénáře*): what a row is, what the claim is about, what is measured, what the null means, the resampling, what the stability measures.
- The stability test, declared before the computation: which alternative settings the map is recomputed under and what agreement counts as stable.
- The method pinned: `aia-sociomap-5`, the organization's methodology in force.

## 9.2 What comes out

- **The futures map** in two views: the 3D terrain and the top view that does not distort distances by perspective. Heights are mean saving; links are drawn only between pairs the contract allows; a warm link means co-movement, its width the relation's strength.
- **The relation matrix:** each pair's relation across the futures, from which the map is drawn.
- **The shared-failure table:** when an option is worse than the baseline, the share of those futures in which each other option is worse too, printed with its denominators (how many bad futures the share is of). This reads the absolute, uncentred futures; the map reads the centred relations, while its heights are the uncentred mean savings.
- **Three separate results, never one word:** layout convergence (did the layout algorithm converge in every setting), representation fit (how well the layout represents the relation matrix) and robustness (the agreement of the distance orders across the declared assumption settings against the threshold). A grouping the report keeps after a failed robustness test is tested on its own, or the matrix is printed without a spatial reading.
- **The register status:** the map's method at its spec fingerprint, approved or *interní* (`domain/sociomap_register.py`), printed on the map and in the report.

## 9.3 Checks

- **Mode contract.** A row is one independent draw of the model. The claim is about the model and its assumptions, not about the market. Centring each future on the mean of the options makes the map show relative performance: who does better in the same future, not whether all do well.
- **Relative apart from absolute.** The question "is option B a backup for A" is answered by the uncentred shared-failure table, not by the map: centring removes the common level, which is exactly what shared failure is made of. The table answers downside dependence only; whether one option could

operationally stand in for another (switching time, cost, feasibility) is a separate assessment the report must name as such. A sentence like "no option is a backup of another" is a claim about tested downside dependence for the options it names, and prints with that scope.

- **Links only where the contract allows.** Two rules compete. The signed-off method's pair test, with the false-discovery correction, marks pairs reliable; the futures mode's contract says a null makes no sense when the model creates the relations, asks for no p-value, and measures stability as the sensitivity of the shape to assumptions declared in advance. Which applies to futures is an open question for the methodologist (chapter 14); until it is answered, the report prints which rule it drew by.
- **Determinedness.** Split-half agreement of two halves of the futures against the method's threshold says the layout is not an artefact of the draw. It does not say the layout is stable under the assumptions.
- **Stability, declared in advance.** The map is recomputed under the alternative settings written down in step 4 (for correlations: none, the judgement's, strong) and counts as stable only if every pair of versions agrees in the order of the inter-option distances at or above the threshold (0,8 by default). A setting whose layout did not converge does not count as one of the settings until it does. Failure is printed, not softened.
- **What the map never claims:** causation; the behaviour of the market or the firm; that a relation is observed rather than a property of the model; that a distance is an effect size in any unit; that the terrain between points is an estimate; validity borrowed from team-communication Sociomaps, which do not cover this use.
- **Internal until approved.** Maps whose observers are futures or properties are an estimate awaiting the methodologist's approval; they print *odhad (metoda čeká na schválení metodikem)* and carry the internal label. The Sociomap of people's ratings is approved since 21. 10. 2026; this is a different mode.

## 9.4 In AIA today

- The `aia-sociomap-5` engine, signed off by the methodologist on 21. 10. 2026 and what a new run pins: `domain/sociomap/signed_off_inputs.py` (per-person centring, the baseline, pair floors, Benjamini–Hochberg), `signed_off_layout.py`, `signed_off_credibility.py` (split-half agreement, map uncertainty), `signed_off_respondents.py`, `signed_off_segments.py`; `engine_v2.py` `compute_object_map` and the contract-3 artefact.
- The register: which method at which spec fingerprint is approved; anything absent is `INTERNAL_ONLY`, never presumed (`domain/sociomap_register.py`, ADR 0038).
- Map snapshots drawn on the server from the stored map at the report's camera and approved as artefacts (`application/map_snapshots.py`, RD4); the map chapters of the report (`application/report_maps.py`, `infrastructure/report_docx/object_map_figure.py`).
- The canvas's own proof that the engine takes a futures table as rows: `prototypes/obchodni-model/tools/engine_maps.py` ran `compute_object_map` on 2 000 futures and drew the maps with the report's renderer; the same tools drew a futures map in three settings and a weighting map.
- The web's map: `ObjectMapPanel`, `lib/object-map-signed-off.ts`, `lib/sociomap-display.ts`, display only.

## 9.5 New

- The futures mode's contract (`sociomapa-systemu#SYS9`): rows a model's draws, each centred on the mean of the objects; the claim about the model and its assumptions, not the market or causes; no p-value; determinedness from the maps of two halves of the futures; stability as the shape's sensitivity to assumptions declared before the computation. It is the first of five readings of one engine (futures,

experts, units, periods, and the weighting map as the fifth). The screen shows the map, the relation matrix, the shared-failure table, the relative and absolute readings side by side, and the stability test with its results and verdict.

- The shared-failure table and the stability test: Spearman on the inter-option distances between every two settings ≥ 0,8, the settings declared before computing, a failure printed as it is (#SYS9b, **new**).
- Each mode its own Metodika entry, internal until approved (#SYS12); the map on Výsledky and in the report (`cost-model-study#CS5`), declared extremes and direction per property (#CS1, reused by the weighting map).
- The weighting-scenario map, a second map beside this one (`sociomapa-systemu#SYS13`, **new**; chapter 12). SI1–SI4 keep only what that map reads: states, gates, coverage, the overlap check.

## 9.6 Acceptance

- The options placed by whether they do well in the same futures; links only where the declared rule allows; the view chosen and the reason stated.
- The map labelled internal until Metodika approves the method.
- The relation matrix and the shared-failure table printed; the backup question answered from the table.
- The stability test declared before the computation, and its result printed whether it passed or failed.
- The map is not part of the recommendation; the matrix, the text and the figure agree.

CHAPTER 10 · STEP 8

# *Osnova a Zpráva*: the outline, then the report

The outline comes before the text. It lists every claim the report will make in the order of the argument, each with its source or method and why it matters to the decision. The report then says only what the outline holds.

## 10.1 What goes in

- The results, the map, the input table, the findings table, the public inputs and the decision record: every artefact of steps 1–7, by id.
- The part structure of a business-decision report and the grade and source conventions (below).
- The validity review of chapter 14, before approval.

## 10.2 What comes out

### The outline (*Osnova zprávy*)

Five parts in the order of the argument: 1 the decision and what is asked of the decision-makers; 2 why the preferred option is acceptable, or why none is; 3 what could change that conclusion; 4 implementation conditions, owners and dates; 5 supporting evidence and technical appendices. Each point carries its source or method, its grade attributes, and one line "why this matters" (*proč na tom záleží*). A judgement is a typed judgement record; a synthesis from sources is an inference record with its evidence. Stops (*zastávky*) close parts for review; a closed stop may go to the client's decision-makers for comment. The screen maps the five parts of the argument onto the report's parts, prints the seven grades with their four attributes, and opens a number's provenance card: source, method, uncertainty, validation, the run, the inputs it stands on, the data check behind it.

### The report (*Zpráva*)

| Part | Rule |
|---|---|
| *Shrnutí*: *Doporučení · Důvod · Jistota · Další rozhodnutí*, the rule table, *Stav podkladů* | one page plus the state box; the rule table without the intervals; the state box names the gaps that could change the result and the public inputs not yet verified (argument part 1) |
| *Úvod*: the situation, the options, the rule | the rule printed in full with its date and the interpretation records |
| *Přehled literatury* | each work with "what follows for this report" |
| *Metodika*: about AIA, the Sociomap, the model in words, the data, the grades | the grade legend is printed in every report; the Sociomap paragraph says which modes are approved |
| *Předpoklady*: the assumptions the verdict rests on | the ones without a document first |
| The result chapters: decision, schedule, levers, information, finance, operations | each chapter's first sentence is its finding; each number carries its grade; sources numbered at the chapter's end (argument parts 2 and 3) |
| The map chapters | relative and absolute readings, the stability result, what the map may and may not say; the weighting-scenario map here, inside argument part 3 |
| *Diskuse* | what the results mean together, and what the model does not resolve |

| Part | Rule |
|---|---|
| *Doporučení a provedení* | a dated plan: when, what, who, output; the enquiries with their owners; what to watch (argument part 4) |
| *Meze studie* | every judgement without a document, every provisional input, every simplification, every method awaiting approval; "each change computed separately" |
| *Literatura, Seznam zkratek* | — |
| *Přílohy* and tables | the document register; the method in detail and the findings that change the result; data checks and computation (seed, N); the tables, generated by the model from the same run (argument part 5) |

Every number carries one of the grades of chapter 13 in small type and a numbered source at the chapter's end; the summary's recommendation status says *předběžné* where the recommendation rests on a method outside the register (chapter 13). The report states its release state on its first page: *Předběžné doporučení* with each unresolved qualification named (unverified public inputs, a decisive judgement without a document, a reading awaiting confirmation, a failed stability test), and *Schválit jako předběžné* as the only approval offered while any remains. The summary may carry a paragraph *Jiné vážení* beside *Jistota*: which options win under the same weights of the perspectives, as a share of tested settings, never a probability, and not part of the recommendation.

## 10.3 Checks

- No point without a source or a method; a point whose source was not captured does not enter the report.
- The report says only what the closed outline holds. A sentence that compares in words with no number behind it is marked (RD7's sentence check, already built).
- Every chapter's title is its finding ("No option yet passes the rule", not "Results").
- Every number is a reference to a stored result: the same run, the same artefact id. Tables are generated by the model from the run, so they agree with the text by construction.
- Counts the report prints (inputs, public inputs, judgements) come from the manifest; a count in the text that the table does not hold is a finding.
- The required disclosures are in every report: the recommendation's status and qualifications, each number's source and method, what is outside the model, who stands behind the assumptions, the method versions.
- The validity review is recorded before approval, and the approval records the exact version and the unresolved qualifications (chapter 14).
- A stop closed with the client's people records their agreement (ADR 0046 W26).
- No respondent, person or e-mail is named beyond a role; the report is Czech in the MVP (ADR 0046 W28).

## 10.4 In AIA today

- The report as data: the document model with sections and blocks, numbers only as refs (`domain/report/model.py`, `document_model.py`); the evidence ledger built from admitted claims with print grades (`evidence.py`); the section-source manifest naming each section's sources (`sources.py`); printed numbers as locked content controls with one tag grammar (`control_tags.py`, `controls.py`).
- The DOCX renderer with deterministic bytes, the Word style sheet from the print tokens, tables with repeating headers and suppressed rows counted, evidence marks as vector images (`infrastructure/report_docx/`); the PDF beside it (`infrastructure/report_pdf.py`).

- Composition: the internal report, the client report, the report with literature, their chapters from verified sources (`application/report.py`, `report_chapters.py`, `report_tables.py`, `report_design.py`, `report_method.py`, `client_report.py`, `contextual_report.py`).
- Approval once per version with the approver recorded and self-approval rules (`application/report_approval.py`, `artifact_repository.approval_record`); the client gate's checks as warnings (`domain/report/client_gate.py`, RD0); the sentence check (`sentence_check.py`, RD7); the report editor with saved versions and their changes against what code wrote (`report_editor*.py`, `saved_changes.py`).
- The stored document read on screen with each number a chip to its claim (`ReportDocumentView`, `lib/workbench-artifacts/report.ts`).

## 10.5 New

- The outline as data with a resolving link on every point (`research-findings-stage#FND0`, `#FND2`), findings in (`#FND3`), tables from points and graphs from tables (`#FND4`, `#FND5`), stops (`#FND8`), *Zpráva* written from the outline (`#FND9`).
- Composition per situation from chapter kinds and outline presets (ADR 0045; `situation-reports#RC1`), fixed method texts signed by the methodology owner (`#RC4`); the business-decision recipe: the parts above as an outline preset with part ids, the grade legend, numbered sources per chapter, *Stav podkladů* and *Literatura*, in the argument order (`research-findings-stage#FND13`, **new**).
- The frame and CALCULATED (`cost-model-study#CP8`), ESTIMATED (`#CP10`), the method appendix (`#CR9`), and the seven print grades mapped to the code's with an admission rule each (`#CP8b`, **new**; chapter 13).
- Approve and export (`report-delivery#RD3`); tables generated from the run into the report (`#FND4`); the limits chapter composed from the input table's states and the run's declared simplifications (part of `#FND13`).
- The validity review before approval: the five questions, extreme-input tests and a named reviewer recorded on the ledger (`research-findings-stage#FND14`, **new**; chapter 14); the release label where the owner's policy decides it.
- The state box and the limits chapter composed from the input table, its counts from the manifest; the *prozatímní* banner on every page while one decisive input is provisional.

## 10.6 Acceptance

- Every number carries its grade and source: file and row or page, or the quoted public source.
- An estimate from a method that is not approved is labelled *odhad (metoda čeká na schválení metodikem)*.
- The report is Czech, as PDF or DOCX, and every part above is present.
- No respondent named or quoted by id.
- Narrative, tables and figures do not contradict one another, percentile and probability wording included; this is checked by code and by a person.

CHAPTER 11

# Built vs designed

No code for business-decision studies is on `develop`. What AIA has is the ground the pivot builds on: the shell, the working copy, client sources, Deep Research, the Sociomap engine and its register, the report document model and its renderer, approvals and audit. This chapter says which is which.

## 11.1 The state on 9 November 2026

| Step | Built on `develop` (reused) | Designed (plan chunk as the scope names it; **bold** = new, in no plan file yet) | Neither yet |
|---|---|---|---|
| *Zadání* | working copy, revisions, edit ops; Brief stage; the Lead; approval ledger; material classes | `decision-brief#DB0`, `DB2`–`DB5`, **DB3b**; `cost-model-study#CP3` (the model inside the one merged study: no new study kind) | the rule freeze with amendments as revisions (chapter 2) |
| *Data* | client sources with readers and locators; dataset preview with personal columns; intake sheet; evidence policy (ADR 0027/0037) | `client-data-in-analysis#2`, `2a`, `3`, `3a`, `3b`, `5`, `10`, `11`, `12`, **`13`**; `cost-model-study#CR1` | — |
| *Rešerše* | Deep Research with grounding and the sealed bundle; RetrievalGate; Wikipedia, Brave, Common Crawl; verified sources | `deep-research-v2#DRV1`–`DRV3`, `DRV5`; `public-data-routes#PDR1`, `PDR2`; `legal-context#LGL1`–`LGL3` | verification at source as a dated act, named in `DRV1`'s state |
| *Předpoklady* | methodology as data (versions, activation, bounds); ledger and audit; the simulation core's typed, validated inputs as a pattern | `cost-model-study#CP1`, `CR2`, `CR3` (mixtures included), **CR3b**, **CR1b**; `client-data-in-analysis#12` | — |
| *Model* | worker, workflows, artefacts with SHA-256, pipeline fingerprints; methodology pins on a run; seeded determinism | `cost-model-study#CP2`, `CP3`, `CP5`, `CP6`, `CM7`, **CM8**, `CR3`, `CP10`; `quantitative-families#FIN1`, `FIN2` | — |
| *Výsledky* | admitted claims and suppression; results reading (unknown never ordered); print grades; table exports | `cost-model-study#CR7`, **CR7b**, `CM2`, `CR4`, `CR6`, **CR6b**, `CR8`, `CP8`, **CM9**; `quantitative-families#FIN3`; `legal-context#LGL3` | — |
| *Sociomapa* | `aia-sociomap-5` engine; the register; map snapshots; report map chapters; the canvas tool that maps futures with the engine | SI1–SI4; `sociomapa-systemu#SYS9`, **SYS9b**, `SYS12`, **SYS13**; `cost-model-study#CS1`, `CS5` | the register entries themselves (the methodologist's) |
| *Osnova, Zpráva* | document model, evidence ledger, control tags, source manifest; DOCX and PDF renderers; composition; approval; client gate; sentence check; report editor | `research-findings-stage#FND0`, `FND2`–`FND5`, `FND8`, `FND9`, **FND13**, **FND14**; `situation-reports#RC1`, `RC4`; `cost-model-study#CP8`, **CP8b**, `CP10`, `CR9`; `report-delivery#RD3` | the release label (the owner's policy) |

## 11.2 The plans that carry the build

From `python tools/progress.py` on 9 November, on `develop`: `cost-model-study` is planned, 0 of 32 chunks; `client-data-in-analysis` in progress, 1 of 15 (8a built); `report-delivery` in progress, 7 of 16; `quantitative-families` and `situation-reports` planned. The other plans the scope cites (`decision-brief`, `deep-research-v2`, `public-data-routes`, `legal-context`, `sociomapa-systemu`, `research-findings-stage`, `analysis-frameworks`, `diagrams`, `register-personal-data`) exist only on branch `chore/pivot-business-decisions`, with ADR 0046, Proposed and not merged; the scope's preamble says so and reads them at `d8dc8ec5`. The chunks the scope marks **new** exist in no plan file at all. ADR 0044

(quantitative models as a study kind) and ADR 0045 (reports composed per situation) are Proposed on `develop`. The scope (§ 2, § 7) maps each step to its chunks and orders them in five waves; this chapter's "Designed" column is that mapping.

## 11.3 Three honest statements

- **No business-decision report has been produced by AIA's product.** A report's sentence "every number was computed by a verifiable program" can be true only of the program that computed it. The MVP's job is to make it true of AIA.
- **The canvas is design.** The boards show the screens and the method; their numbers are fictional and illustrative, not validation evidence. The boards hold the method; the plan chunks hold the build.
- **The methods of the futures are not registered.** Monte Carlo with PERT and correlations, the decision boundary, both measures of the value of information, the futures map and the weighting-scenario map await the methodologist's entries (chapter 14). Until then their results print *odhad (metoda čeká na schválení metodikem)*, and a recommendation that rests on them is *předběžné* under the release policy the owner adopted on 9 November (chapter 2).

## 11.4 Not in this release

Cut from the ten weeks on 9 November, to keep the spine to what the first study type needs: the framework catalogue and picker in *Zadání*; model types other than `site_cost_of_supply`; automatic public-data and legal connectors (public inputs stay typed and sourced, entered by hand); fitted demand forecasting; global sensitivity by Sobol or PRCC; the respondent, expert, unit and period map modes; survey processing; the generic diagram editor; the shared client-review interface; monitoring. Each stays in its plan file as parked.

CHAPTER 12

# Perspectives, the cost partition and the two maps

The owner asked for every business perspective: financial, strategic, operational, marketing, logistics, "just everything", with weights on them run as scenarios and shown on the Sociomap. The method gives that request one construction: every pre-tax cash-flow line of an option is allocated to exactly one leaf under exactly one perspective, a perspective's total is the discounted cost of its leaves, and one value function and one weight vector score every option alike. Overlap is impossible by construction, and "logistics 0,3, operations 0,5" means exactly what it says.

## 12.1 The partition

A **leaf** is one cash-flow line (or a group of lines that always travel together), discounted at the rule's rate. A **perspective** is a set of leaves. The same expenditure never appears in two leaves; distinct expenditures caused by the same event may. Delay costs enter through the purchases they actually cause, not as a separate penalty. The allocation for a cost-of-supply study:

| Perspective | Leaves (each a cash-flow line, discounted) |
|---|---|
| *Finanční* | core capital; main-plot land; working capital where applicable; the purchases that continue until start-up, ordinary and spot; residual credits and other financial adjustments not allocated below |
| *Strategická* | no scored leaf until an independently defined consequence is approved (an open owner decision). The shared-failure analysis of chapter 9 stays a table of its own, never a weighted leaf. |
| *Provozní* | raw materials; electricity and gas; maintenance and insurance, excluding the services allocated below |
| *Marketingová / obchodní* | no scored leaf where the output is used in-house; the perspective is drawn and says why it is empty |
| *Logistická* | product freight; connecting infrastructure and its land; access works; temporary haulage until that infrastructure is built |
| *Prostředí a povolování* | treatment capital and operation; air abatement; compensation measures; water and sewer; hazardous waste; carbon (allowances less free allocation); separately identified environmental and site services |
| *Lidé* | wages, employer contributions and evidenced recruitment expenditure. Hiring time enters the timetable and therefore the gates, not a second weighted penalty. |

Three checks hold the allocation: every expenditure the documents name is allocated once; the perspective totals sum to the option's cost of supply; the baseline's cost less the option's cost equals the NPV saving. A one-off the documents give for one option only goes into the leaf its nature places it in (a bridge in logistics, a compensation measure in environment and permitting); the other options carry an evidenced zero there, or a justified range where the documents are silent. An expense that fits no leaf blocks the reconciliation check: the partition is then not complete, the map is not drawn from it, and the report says which expense is unplaced.

Five rules hold the scores:

- **Gates are not leaves.** The rule's conditions are pass / fail / unknown per option, shown beside the map, never weighted.

- **A range stays a range.** A leaf the files give as a range has a lower and an upper score bound, the direction applied correctly: for a cost the minimum raw value is the *favourable* end, so the map never draws it as the base. Both bound views are shown and labelled as bounds, not as jointly feasible futures. Nothing fills a missing fact, and a fact the client's file states is not widened into a range.
- **One expenditure, one leaf.** NPV saving, after-tax NPV, IRR and payback are printed beside the weighted score, unweighted, because they already contain every leaf.
- **Judgement is labelled.** A leaf whose value is a person's judgement carries that person's name and the grade *úsudek*.
- **An empty perspective stays and says why**; a globally inactive perspective has weight zero for every option, approved as such. A nonzero weight on an empty perspective is refused with the reason; nothing is redistributed silently.

## 12.2 One scoring structure for every option

For an option *i* and a perspective *p*, $C_{ip}$ is the discounted sum of *p*'s leaves. The approved value function $v_p$ maps it onto 0–10 on fixed anchors shared by all options; a value outside the anchors triggers a review, never silent clipping. The weighted score is

$$S_i(w) = \Sigma_p \, w_p \cdot v_p(C_{ip})$$

- **One weight vector for every option.** No option loses a weight because a leaf does not apply to it: evidenced absence of an applicable expenditure is a cost of zero through the common scale; unknown expenditure is a range with two bounds. A rule under which a missing component "takes no weight there" would compare options under different trade-offs, and is not used.
- **Leaves are an accounting breakdown, not equally weighted subcriteria.** Splitting a cost line (freight into rail and road) leaves the perspective total, the score and the rank unchanged. This is the admission test of every allocation; sharing a perspective's weight equally among its components fails it.
- **Anchors and swing weights are approved together**, before any option's score is shown: a weight is a trade-off over a stated swing of a perspective's total, not a feeling about a category name. The additive form itself is reviewed with them.
- **A scenario is a named weight vector** the client's people approve; the researcher records it with its scales through the existing approval flow (the owner, 9 November), with who composed it and when, before results.

An example, with the remaining weight assigned to finance: *finance 0,20 · strategie 0 · provoz 0,50 · marketing 0 · logistika 0,30 · prostředí 0 · lidé 0*. "Finance" is its partition, the capital, land and purchases, not the total NPV. The NPV still includes every cost, including the perspectives given zero diagnostic weight.

## 12.3 Ranges and sensitivity

A ranged leaf gives the option an interval of scores. A **robust leader** needs a validated positive lower bound on its score difference against every rival over all admitted unknowns jointly; overlapping intervals mean leadership is not established, and the map says so. For each named scenario the study tests, in this order and reported separately:

1. the approved centre;
2. Dirichlet fans at concentrations 20, 60 and 180, 300 settings each, drawn on the positive-weight coordinates only so an explicit zero stays zero;

3. directed weight transfers between pairs of perspectives, and a simplex grid where the number of active perspectives keeps it small;
4. the approved alternative anchors and the uncertain cost allocations;
5. the evidence bounds, coherent joint stress cases and simultaneous unknowns.

Results are shares of tested settings, per scenario, per fan and per evidence setting: the share in which an option is uniquely first, the share of ties and the share left unresolved by the bounds, which together reconcile to 100 %. Scenarios are never pooled into one popularity figure.

## 12.4 The Sociomap view of a scenario

Each scenario has its own view on the `aia-sociomap-5` engine, under the mode contract the methodologist registers for it (`sociomapa-systemu#`SYS12, SYS13):

- rows are the tested weight settings; objects are the options; proximity is similarity of score response across the settings; height is the score under the selected setting;
- a ranged leaf keeps its paired bound views and score intervals in the table beside the map;
- the gate status of every option (pass / fail / unknown) is printed beside it; a hypothetical change of a gate's state is a named scenario, never an edit;
- the rule's verdict stands in a panel of its own; the map is preference information under *Co by závěr změnilo* and never the recommendation;
- no respondent labels, no "reliable pairs", no respondent segments and no sampling-error circles appear: those belong to the respondent mode, whose statistics this mode does not read;
- the view records its layout fit, its numerical convergence and its sensitivity under the mode contract; geometry the contract does not support produces no spatial claim.

The canvas's weighting-map board predates this construction; it is recomputed under the partition at the next canvas version (the owner, 9 November) and carries a note saying so until then. The other map, the futures map, and its contract are in chapter 9.

CHAPTER 13

# Grades, sources and the release gate

Every number in a business-decision report carries a grade in small type and a numbered source at the end of its chapter. The grade says how the number came to be; the source says where. Behind the one printed grade stand three more attributes.

## 13.1 The seven grades

| Grade | Meaning | Typical instance |
|---|---|---|
| *z dat klienta* | a number read from a client's file or document | a contract's end date; the comparables' overruns |
| *výpočet* | a number the program computed from stated inputs in one computation: the base case, break-even, finance, operations | the base-case saving; the finance and operations tables |
| *odhad* | a range derived from client data, for instance from comparables; and every number from the futures, which carries the full label *odhad (metoda čeká na schválení metodikem)* while the method is unregistered | every median, interval, probability, value of information, boundary, delay table and the futures map |
| *úsudek* | the author's assumption where the client has no answer; also proposed dates | the chance of an event no document prices; a judged correlation; the plan's dates |
| *veřejný zdroj* | a figure from public research, provisional until verified at the source | a market price; exchange rates |
| *právní předpis* | a provision, likewise provisional until verified | tax rates, employer contributions, regulatory limits |
| *průzkum* | answers from a survey; a business-decision study does not use it | — |

## 13.2 The printed grade and the three attributes behind it

The printed grade says how the number came to be: its origin (*z dat klienta*, *veřejný zdroj*, *právní předpis*, *úsudek*, *průzkum*) or, for a model result, its method (*výpočet*, *odhad*). Three more attributes travel with every number and are never folded into the grade: its **lineage** (which inputs it was computed from and their grades), its **uncertainty** (the range, interval or probability basis, printed beside it) and its **validation** (the method in the register or awaiting the methodologist; a public input verified or *prozatímní*). A base-case figure computed from a judgement therefore prints *výpočet*, and its lineage shows *úsudek*, which the sentence must state where the judgement is decisive.

| Printed grade | Who may print it | Admission rule (code) | Carried beside it |
|---|---|---|---|
| *z dat klienta* | a number read from a client file or document under its contract | CLIENT_REPORTED: the source locator resolves to the accepted contract and copy | validation *ověřeno*; lineage the file and row |
| *výpočet* | one deterministic computation from stated inputs: base case, break-even, finance, operations | CALCULATED: the run's artefact and the inputs' revision pinned | lineage: the inputs and the weakest grade among them; uncertainty: none of its own |

| Printed grade | Who may print it | Admission rule (code) | Carried beside it |
|---|---|---|---|
| *odhad* | a statistic of the futures, or a range derived from client data | ESTIMATED: the method's register status and the comparison's resolution state | validation: *(metoda čeká na schválení metodikem)* while unregistered; uncertainty: the interval, and *nerozhodnuto* where the bound crosses a threshold |
| *úsudek* | a person's assumption where the client has no answer | a labelled judgement with its author's name | lineage: none; uncertainty: the range the author stands behind |
| *veřejný zdroj* | a figure from public research | a typed input with quote, source and date | validation: *ověřeno* or *prozatímní* |
| *právní předpis* | a provision | a typed legal input (`legal-context#LGL5`) | validation as above; the effective date |
| *průzkum* | survey answers; not used here | the survey origins | — |

In the code the print grades are five today, `PrintGrade` in `domain/report/evidence.py`: measured, calibrated, modelled, holdout-pending, unknown, a missing grade printing "?". The plan adds CALCULATED and ESTIMATED with the register's status (`cost-model-study#CP8`, `#CP10`), ADR 0027 the client-reported origin, and `#CP8b` (**new**) the table above as code: one row per grade, Czech label, admission rule, the attributes carried.

## 13.3 Sources, numbered per chapter

- Each chapter ends with *Zdroje a stupně*: a numbered list; each printed number cites a list entry by its number ("z dat klienta 1").
- An entry names the document by its cited name and page or paragraph ("Zápis zadání s klientem §6", "Dopis dodavatele, s. 1"), the file for data, the table for a computed value ("výpočet: tabulka v příloze"), or the method's status for an estimate ("odhad (metoda čeká na schválení metodikem)").
- An entry may carry two grades when a number rests on both ("výpočet; z dat klienta: srovnatelné projekty"); the weaker is printed at the number.
- The document register maps every cited name to its file, so a source is checkable with one open.
- In AIA, the section-source manifest (`domain/report/sources.py`) already records each section's sources by artefact id and SHA-256; the numbered list is its printed form.

## 13.4 The provisional state

*Prozatímní* is a validation status, not a grade. It marks a public input not yet verified at the source, and it is printed in three places: the input table's state column, the sentence that uses the input ("dlouhodobý vývoj cen energie (prozatímní údaj)"), and the first-page state box with the count. The count comes from the manifest (chapter 2), never from the text. It is removed only by a dated verification by a named person. The report's running head carries *Pracovní verze · prozatímní* on every page while any decisive input is provisional.

## 13.5 The release gate (adopted 9 November 2026)

The owner adopted the release policy on 9 November: labelled unregistered estimates may appear internally and in an independently reviewed *předběžné* client draft; an approved recommendation needs registered methods, the independent review and every decisive comparison resolved. Document approval

and recommendation eligibility are two states (chapter 2). This amends W27 (an unregistered estimate reaches the client report labelled, with a warning) and report-delivery RD0 of 22 October ("warnings for all of it") for the business-decision report: the warnings stay, and the recommendation's eligibility is a computed state beside them. The amendment is recorded in ADR 0046 by the docs PR that follows the scope's version 2.

- Every decision-critical claim is checked twice: support in its source (code) and correct interpretation (a person who is not the claim's author).
- A recommendation that rests on a method outside the register, or on a comparison still unresolved under the convergence rule, is *předběžné doporučení*, and its summary says so.
- Exploratory material enters the report only with explicit limits ("exploratory; does not support the recommendation"). The weighting-scenario map is such material.
- Approval records the exact version and the open qualifications; "approved document" means reviewed and releasable, "recommendation eligible" means the recommendation may be stated as approved. The second is never typed; code derives it from the register, the review record and the resolution states.
- The Deep Research thresholds (citation precision ≥ 90 %, unsupported claims < 10 %) are one check, not the release criterion.

What exists today: approval once per stored version with the approver and self-approval rules recorded (`application/report_approval.py`); the client gate's checks as warnings recorded on the approval's ledger row (`domain/report/client_gate.py`, RD0); the sentence check (RD7). What the policy adds: the recommendation's eligibility as a computed field of the approval, the qualifications list, and the second, human reading of each decision-critical claim.

CHAPTER 14

# Governance

Reproducibility proves the code computes what is written. It does not prove the model is the right one for the decision. That needs a review by someone who did not accept the inputs, a register that says which methods are approved, and a record of what is still open and whose word closes it.

## 14.1 The validity review

Before a recommendation is approved, an independent reviewer answers five questions; the researcher who accepted the inputs answers to them, never reviews them.

| # | Question | What the answer rests on |
|---|---|---|
| 1 | Is every option represented comparably? | the same scope, horizon, rate and timeline; option-specific months and costs from the option's own documents |
| 2 | Are the material costs, benefits, constraints and implementation dependencies in the model? | the scope check of every supplied estimate; the list of what is outside the model, stated where the model is described |
| 3 | Could an excluded item reverse the recommendation? | each exclusion's size against the margin of the verdict; an excluded item that could reverse it named on page 1 |
| 4 | Do extreme inputs give sensible results? | the extreme-input tests run: each decisive judgement at the ends of its plausible range, the decisive price at an earlier level, each alternative reading of a condition, a delay with the decisive event ruled out; each behaving as expected |
| 5 | Has an independent reviewer challenged the structure and the decisive assumptions? | a named reviewer who did not work on the study, the date assigned, and the questions still open |

With the questions, the review holds a table of the limits the report admits, each with the check that catches it and its state, caught or open; the statement that validity is not reproducibility; and the six review criteria with their state: comparable options with material costs and dependencies in the model; uncertainty fit for the decision; the rule's choice kept apart from supplementary scoring; a revised input propagating to everything that read it; every enquiry stating its effect on the deadline; and claim integrity between narrative, tables and figures. The review's record is part of the approval; its open items are the qualifications that keep the recommendation *předběžné*.

## 14.2 Register entries the method waits on

The Sociomap register (`domain/sociomap_register.py`) and Metodika's versions (`domain/methodology/`, ADR 0023) hold what is approved. A business-decision study needs entries that do not exist. Until each exists, its results print *odhad (metoda čeká na schválení metodikem)*. NPV, IRR, payback, break-even and the tornado stay *v registru*: a single computation needs no entry.

| Method | What the entry must fix | Where it is used |
|---|---|---|
| Monte Carlo over the input table: PERT, declared correlations, empirical joint draws from comparables, yes/no events, mixtures | the default distribution by source; the correlation method; the stopping rule and its error budget; the seed convention | every median, interval and probability |
| The decision boundary | the grid over an input's range; recomputation of the futures per value; how the flip is reported | the decision view |
| Value of information for the rule, and EVPPI | the partition of futures by the answer; the rule applied per partition; EVPPI by Strong and Oakley (2013); the order of the two measures; the enquiry table's columns | the information view |
| The futures map (observers = futures) under `aia-sociomap-5` | the mode contract: which pair statistic, if any; the determinedness test; the stability test's settings, agreement measure and threshold; the relative and absolute readings (`sociomapa-systemu#SYS9`, `SYS9b`, `SYS12`) | the map chapter, the relation matrix, the shared-failure table |
| The weighting-scenario map | the partition's admission test (splitting invariance), value anchors and swing weights, the fans' concentrations and k, score bounds and the robust-leader rule, shares of tested settings per scenario, the view's mode contract (`sociomapa-systemu#SYS13`, `SYS12`) | the weighting chapter |
| Estimates from comparables | the filter, the statistic, the minimum sample, how a joint draw is recorded (`cost-model-study#CR3b`; `client-data-in-analysis#12`, an estimate from client rows as a Metodika method) | overrun and delay; ranges of hidden site costs |

## 14.3 The methodologist's open questions

ADR 0046 lists ten (§ Open: multiple testing on maps; levels or changes; effective sample size; resampling for 24–36 rows; the surrogate null; the relation statistic; the synthetic suite's thresholds; the discount rate in probabilistic rules; PERT inputs; forecast evaluation), each with a recommendation. This method adds nine:

1. **The futures map's contract** (`sociomapa-systemu#SYS9`): the rows are chosen draws, so what does it print instead of the respondent method's count of reliable pairs and its split-half agreement? The signed-off method draws links by its pair test with the false-discovery correction; the futures contract says no null applies when the model creates the relations.
2. **The stability test**: Spearman on the inter-option distances between every two of three correlation settings, threshold 0,8; approve, or replace the statistic and the threshold. The "strong" setting is the author's judgement and must be declared as one.
3. **The empirical joint draw** (`cost-model-study#CR3b`): one comparable per future carrying overrun and delay together, as an alternative to PERT with a declared correlation; the minimum number of comparables that supports it, and how outliers are treated.
4. **The convergence rule and the endpoint semantics of chapter 2**, adopted as the baseline on 9 November: confirm or amend the doubling from 20 000 to 1,28 million, the repeated-look exact binomial bounds, the mean bound for information comparisons, the 5 % error budget, quantiles fitted as quantiles and hard bounds as support; PERT λ; and when a mixture or a uniform is the better default for inputs the files do not hold.
5. **Dependence**: the contract recommends a Gaussian copula for continuous inputs (validated rank-to-latent conversion, a valid latent correlation matrix, preserved empirical pairs, explicit event dependencies) in place of sample-wide Iman–Conover reordering, so the numerical error contract

matches the sampler; confirm, and set the tail stresses and the achieved-dependence check.

6. **Value of information for the rule** (`cost-model-study#CR6b`): the rule re-applied per declared answer branch with the enquiry's cost, duration and shared dependencies inside the comparison; approve the definition, including its meaning when the rule's choice is "nothing", how it combines across several enquiries, and the estimator checks on the fixture O-INFO (17,5 for the rule, 12,5 as expected-benefit EVPI; § 14.6).
7. **The weighting map under the partition** (chapter 12): the value functions' anchors per perspective, the swing-weight elicitation, the fans at concentrations 20, 60 and 180 with 300 settings each, the robust-leader rule on score intervals, and the map-fit thresholds of its mode contract; shares are shares of tested settings, per scenario, never pooled.
8. **The seven print grades mapped to the code's** (`cost-model-study#CP8b`): one table, Czech labels, admission rules.
9. **A register entry** for `aia-sociomap-5` with futures or settings as observers (today's entry covers respondents' ratings), or a separate method id per mode (`sociomapa-systemu#SYS12`).

## 14.4 The owner's decisions of 9 November, and what is still open

| Decision | Where it stands |
|---|---|
| A condition that can be read two ways has its reading fixed and recorded before any result; the test case reads its investment ceiling as total base-case installed investment. | decided; chapter 2 |
| The comparables' cost basis: a versioned clarification of the source on a like-for-like basis, or two bounded interpretations with conditional conclusions. | **open**; recommended: the clarification (chapter 6) |
| The perspective weights weigh a disjoint cost partition, one weight vector for all options, NPV printed beside. | decided; chapter 12 |
| A perspective with no leaf in a study (strategic, marketing): shown at weight 0, or researched for evidenced objectives first. | **open**; recommended: weight 0 until a consequence is defined |
| The release policy: qualified *předběžné* drafts after independent review; approved recommendations only on registered methods and resolved comparisons; two states. | decided; chapters 2 and 13; the ADR 0046 amendment follows in the docs PR |
| The researcher records client-approved weight vectors through the existing approval flow. | decided |
| Three build sessions, one analyst session that never writes model code, the methodologist one day a week. | decided; the weekly day is asked of the methodologist |
| The seven contract defaults as the baseline for the methodologist's review. | decided; chapter 2 |
| Acceptance: the independent numerical oracle first, the frozen test case second. | decided; § 14.6 |
| A separate session writes the oracle; the methodologist signs the fixtures. | decided |
| The attributes of a number as the vocabulary of the outline and the report. | the mapping table is `CP8b` (chapter 13) |
| Enforce data classes before a client's data flows into a model. | yes for Class A in staging and production, recommended |
| The SOMECS name and licence. | until checked, the map is *Sociomapa systémů* |

Settled by the scope: the MVP's spine is the business-decision study end to end; its model is a model type of its own, `site_cost_of_supply` (`cost-model-study#CM8`), inside the one merged study (`#CP3`: no new study kind). The accounting and market result views stay on the canvas as parked.

## 14.5 Change propagation

A revised input is a new revision of the input table. Every result, table, map and approval that read the old revision is marked stale until someone reviews it; the report's version carries the revision it was composed from. In AIA today the pipeline's fingerprints and the impact rule (`domain/pipeline.py`, the research impact route) do this for a research design; the model platform reuses the rule (`cost-model-study` CP5: inputs frozen by artefact id and SHA-256).

## 14.6 Acceptance: the oracle first, the test case second

A pipeline could pass a test case with wrong cash flows: find the planted facts, add a charge for each hidden cost, apply the rule faithfully to incorrect numbers, print the caveats, reproduce the same bytes. So acceptance has two layers (the owner, 9 November). An analyst session that never writes the model code, and never imports it, freezes equations, inputs, expected intermediates, tolerances and conventions; the methodologist signs them.

| Fixture | Content | Expected |
|---|---|---|
| **O-CF** | two years, end-year flows, 10 % discount, no inflation, tax, residual or uncertainty, toy units. Buying: 0 capital, 100 a year. A: 50 capital, 60 a year. B: 20 capital, 80 a year. $C = I + c_1/1{,}1 + c_2/1{,}1^2$; saving = $C_{buy} - C_{option}$. | present cost 173,554 / 154,132 / 158,843; saving 0 / 19,421 / 14,711; at zero discount 30 and 20; a terminal receipt of 12,1 at year 2 lowers A's present cost by exactly 10,000. A separate dated-flow fixture tests the study's own convention (mid-year, inflation); this toy does not prescribe it. |
| **O-INFO** | two equally likely states; A saves 40 / −20, B saves 5 / 5, buying 0; start and investment gates pass; P(beats buying) ≥ 0,75 and P(loss > 20) ≤ 0,10 as gates. | without information the rule picks B (A fails the gate at 0,50): value 5. Perfect information: A in the good state, B in the bad: 22,5. Value for the rule 17,5. Expected-benefit optimisation picks A at 10, so its perfect-information value is 12,5. Both are reproduced and told apart. |
| **Frozen test configuration** | the test case frozen after the cost-basis reconciliation: file hashes, transformations, the manifest, equations, event precedence, the shared and option-specific shock structure, the rule and the method versions; investment, timing and cash-flow checkpoints computed independently before production results are compared. | where the case can be read two defensible ways, each reading frozen as its own configuration; each configuration's numbers a target for itself, never for the other. |

The tests the fixtures feed:

- **Units, currency and periods:** a kilogram row normalised to tonnes by hand, a foreign-currency row at a frozen dated rate, a deliberately misclassified quote; silent substitution fails.
- **Accounting identities and scope:** every expense once; zero usable tax capacity gives no immediate tax benefit.
- **Criterion splitting:** a logistics line split with its total conserved leaves scores, ranks and shares unchanged; scoring parent and children together fails admission.
- **Missing-data states:** known, evidenced zero, globally irrelevant, unknown range, several unknowns that jointly reverse a ranking; no option-specific reweighting.
- **Threshold uncertainty:** known probabilities below, at and above a threshold; samples around 0,79 and 0,81 against 0,80; the whole stopping procedure under repeated seeds, not one final interval.
- **Correlated information branches:** a finite joint-state table with impossible answer combinations; an independent-marginal substitution fails.

- **Distributions and dependence:** analytic PERT moments, quantile-fitted marginals, paired observations, common regional and option-specific shocks.
- **Maps:** known matrices, identical profiles, a dominated option with a similar profile, poor geometry, deliberate instability; similarity never becomes a co-winning claim.
- **Downstream invalidation:** every stored root mutated, descendants stale, old artefacts immutable.
- **Claim meaning and release:** wrong percentile wording, rounding that hides a gate failure, an unregistered method, an unresolved comparison stated as a verdict.

The integration exit is one real browser, API and worker journey on the test case's files that produces every results table, both maps, the enquiry table and the QA annex from frozen artefacts and passes the independent review.

## 14.7 The independent review's evidence

Recommendation eligibility (chapter 2) needs a review record, not a reviewer's name. The record holds: the five questions of § 14.1 answered with the evidence each rests on; the limits the report admits with the check that caught each or the reason it is still open; the extreme-input tests actually run (the register entry says how many a study must run and which; `FND14` builds four); the oracle fixtures' pass record with the signer's name; the resolution state of every decisive comparison; and the six review criteria each marked met, partial or open with its anchor. A review with an open item leaves the document approvable as *předběžné* and the recommendation ineligible.

APPENDIX A

# Glossary

| Term | Meaning in this methodology |
|---|---|
| Base case (*základní případ*) | one computation with every input at its likely value; compares options and feeds sensitivity; not an estimate of the saving |
| Baseline | the option every other is compared with, often the status quo (keep buying, keep the current supplier) |
| Leaf | one cash-flow line (or lines that always travel together), discounted, allocated to exactly one perspective; a perspective's total is the sum of its leaves |
| Data contract (*datová smlouva*) | which columns of a client file are used and how; units, currency, price year, personal columns, period; accepted by a person |
| Decision boundary (*hranice rozhodnutí*) | what the rule chooses as one assumption moves over its range, the futures recomputed for each value |
| Disposition | what was done with a data finding: used; corrected with the reason; left out; taken as an assumption with a range |
| EVPPI | expected value of partial perfect information: the gain in mean outcome from knowing one input before deciding (Strong and Oakley 2013) |
| Fan | k perturbed weight vectors around a named weighting scenario, from a declared concentration; a sensitivity setting, not a probability |
| Futures (*budoucnosti*) | the seeded draws of all inputs at once; probabilities are shares of them |
| Futures map (*mapa budoucností*) | a Sociomap whose observers are the futures and whose objects are the options; centred per future |
| Gate | a non-compensable requirement; fail or unknown blocks the recommendation |
| Grade (*stupeň*) | the small-type label on a printed number: how it came to be; chapter 13 |
| Guarantor (*stojí za ním*) | the person who stands behind an input's range |
| Interpretation record | a reading of a rule's condition that the rule does not fix, with the alternative, its effect and who confirms |
| Mode contract | per Sociomap mode: what a row is, what the claim is about, what is measured, what the null means, the resampling, what the stability measures |
| Oracle fixture | a case with independently known answers (O-CF, O-INFO and the frozen test configuration) frozen by an analyst who does not write the model code and signed by the methodologist; the pipeline must reproduce it before any test case counts |
| Outline (*Osnova zprávy*) | the report's claims in the order of the argument, each with its source and why it matters, before any text |
| Perspective | a set of leaves: financial, strategic, operational, marketing/commercial, logistics, environment and permitting, people; one weight per perspective, the same for every option |
| Recommendation eligibility | the computed state that lets a recommendation be stated as approved: registered methods, the independent review closed, every decisive comparison resolved; apart from document approval |
| Repeated-look bound | a confidence bound still valid after the draws have been doubled and checked several times; the convergence rule uses exact binomial bounds of this kind |
| Provisional (*prozatímní*) | a public input not yet verified at the source |
| Register (*registr metod*) | the methods the methodologist has approved, each at its version and fingerprint |
| Rule (*pravidlo*) | the client's decision rule, written before any result: the measure, the choice, the conditions with thresholds |

| Term | Meaning in this methodology |
|---|---|
| Shared failure | when one option is worse than the baseline, the share of those futures in which another is worse too, printed with its denominators |
| Share of tested settings | the share of a fan's weight vectors under which an option is uniquely first, with the shares of ties and of unresolved settings beside it; per scenario, never pooled; not a probability |
| Stability test | the map recomputed under settings declared in advance; stable only if the distance orders agree above the threshold |
| Stopping rule (convergence) | how many futures to run: 20 000, doubled at declared checkpoints up to 1,28 million, until every gate and decisive comparison clears its repeated-look bound; a comparison that does not prints as unresolved |
| Unresolved (*nerozhodnuto*) | a comparison whose bound still crosses its threshold at the draws run so far; printed as an interval, never as pass or fail |
| Value for the rule (*hodnota pro pravidlo*) | what knowing an answer in advance and applying the rule again would bring, averaged over the possible answers |
| Weighting-scenario map (*mapa scénářů vah*) | a Sociomap per scenario whose rows are tested weight settings and whose objects are the options; height is the score under the selected setting; preference information, with the gates beside it and the rule's verdict in its own panel |